El Mehdi CHOUHAM

Machine Learning Engineer - NLP

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https://scholar.google.com/citations?user=HCWCcIIAAAAJ&hl

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Education

Engineering Degree - Master's
Ecole Centrale de Lille
Sep 2018 – Sep 2021 | Paris, France
Machine Learning / Deep Learning / Reinforcement Learning / NLP / computer vision / Vector Databases / Project Management / Linux / Statistics / Data management / Problem solving

Preparatory Classes
Jules Ferry Preparatory Classes
Sep 2016 – Jul 2018 | Paris, France
Algebra - Statistics and Probability / Differential Calculus & Optimization

Skills

Tools
Python, SQL, Typescript

Deep Learning
PyTorch, Tensorflow, Triton, JAX, Transformers/ Hugging Face, Onnx

Stack
Langchain, Unsloth, VLLM, TGI, AWS Bedrock, Sagemaker, Vertex, AI

MLOps
Docker, Git, Terraform, FastAPI, Langfuse, AWS, GCP, W&B, Weave

Vector Databases
Postgres/pg-vector, Milvus, MongoDB, Chroma, Pinecone

Publications

GPT-3.5, GPT-4, or BARD? Evaluating LLMs reasoning ability in zero-shot setting and performance boosting through prompts
J.L. Espejel, E.H. Ettifouri, M.S.Y. Alassan, E.M. Chouham, W. Dahhane
Natural Language Processing Journal, 2023 (176 citations)

A comprehensive review of state-of-the-art methods for Java code generation from natural language text
J.L. Espejel, M.S.Y. Alassan, E.M. Chouham, W. Dahhane, E.H. Ettifouri
Natural Language Processing Journal, 2023 (25 citations)

Professional Experience

Machine Learning Engineer - Agentic (Freelance) Jan 2026 – Present
Sanofi Paris, France
Built an agentic system to automate patient case reviews in clinical trials, capturing safety narratives for events of interest and cross-checking them against hospital forms and pharmacovigilance data to flag discrepancies.

- Designed an agentic flow (**Pydantic AI** + **Bedrock**) that extracts safety narratives from clinical trial events and reconciles them against all related hospital forms and pharmacovigilance records
- Adopted an eval-first approach: validated core use cases against data labeled by clinical users cleared on the studies of interest (**F2 90%**)
- Built a feedback loop to capture, integrate, and train on reviewer corrections, with prompt optimization via **GEPA** (**DSPy**)
- Stood up eval infrastructure on **W&B Weave** to track quality across iterations
- Increased throughput **~4x** in case reviews per user

Stack: Python, Pydantic AI, AWS AgentCore, Bedrock, DSPy, W&B Weave

Lead Machine Learning Engineer - NLP (Freelance) Apr 2024 – Dec 2025
SNCF Connect & Tech Paris, France

Led **NLP evolution** and **generative AI integration** for France's two largest customer-facing railway chatbots.

- Migrated **MaxEnt** intent/entity models to **transformer-based architecture** (**+13% F1**)
- Cut **inference cost by 74%** through model and infrastructure optimization *(separate team)*
- Designed and deployed a **RAG search agent** with rephrasing and memory capabilities using **Claude/Cohere** on **Bedrock + MongoDB** (**96% recall @k=2**)
- Led **synthetic data generation**, annotation, and traffic simulations pre-deployment
- Reduced bot **response latency by 95%** through tight **Kotlin** integration, replacing microservice architecture

Stack: Python, PyTorch, AWS Bedrock, MongoDB, LangChain, Terraform, Kotlin

Machine Learning Engineer - NLP Jan 2024 – Apr 2024
Apicil - Novelis Paris, France

Sole **NLP/ LLM engineer** tasked with building a **RAG agent for insurance brokers**, helping them find answers and present offers.

- Built a **PDF ingestion pipeline** with parsing, chunking, and **hybrid embedding** (**91% recall @k=2**)
- Set up **Mistral** with engineered prompts, prototyped with **LangChain**
- Deployed the **RAG agent** into production via **FastAPI**
- Conducted business interviews to build a **corpus of frequent queries** and evaluation test set

Stack: Python, PyTorch, Mistral, LangChain, FastAPI

Research Machine Learning Engineer - NLP Jan 2023 – Apr 2024
IRIS - Novelis Paris, France

Built a **Computer use AI agent** enabling users to automate web and Windows OS tasks through natural language.

- Designed the **end-to-end agentic pipeline**, from user intent to executable system commands
- Fine-tuned and quantized an **LLM for instruction decomposition** into elementary actions

- Built a **Mixture of Experts** tool (**transformer** on **spaCy** + **SVM**) for action type classification and execution (**+12% recall**)
- Built **speech-to-text** (Whisper) and multilingual translation (FR/ES → EN) input modules
- Deployed microservices on **GCP** via **Docker** + **FastAPI**, orchestrated with bash/crontabs

Stack: Python, PyTorch, spaCy, Whisper, GCP, Docker, FastAPI, Terraform

Machine Learning Engineer - NLP May 2023 – Sep 2023
Conseil d'État - Novelis Paris, France
Delivered and **integrated an NLP solution** into a French public institution, modernizing legal document processing pipelines.

- Built and deployed a **RAG agent** (Retriever + Reranker + Mistral 7B, **ReAct**) for information extraction from legal PDFs
- Fine-tuned **BLOOM** for multi-page legal document summarization
- Implemented **OCR + PII anonymization** on legal submissions (Tesseract)
- Benchmarked against **GPT-4** baseline

Stack: Python, PyTorch, Mistral 7B, BLOOM, Oracle, Tesseract, Streamlit

Research Machine Learning Engineer - NLP Oct 2021 – Dec 2022
AIDA - Novelis Paris, France
Built a **Java code generation copilot** from natural language project descriptions, involving extensive research, ablations, and iterative experimentation across model architectures.

- Fine-tuned **CodeBERT** for function name generation from docstrings
- Pre-trained **CodeT5** from scratch using **MSP obfuscation** techniques
- Trained a **spaCy NER pipeline** for entity extraction to code classes
- Conducted **extensive ablations** across architectures and training strategies before achieving benchmark-winning results
- Ranked **1st and 2nd on Microsoft CodeXGLUE** benchmark (JacoText)
- Co-authored 2 publications on **LLM evaluation** and **code generation** (176 + 25 citations)

Stack: Python, PyTorch, CodeBERT, CodeT5, spaCy, BigQuery

Data Scientist Oct 2021 – Dec 2021
Lussier - Novelis Paris, France
Analyzed company data and built decision-support tools for a data-driven transformation initiative.

- Built a **metrics dashboard** for continuous monitoring of company performance (Plotly)
- Explored feasibility of a **predictive risk model** using Scikit-Learn
- Developed **data pipelines** for efficient information flow

Stack: Python, SQL, Scikit-Learn, Plotly

Research Machine Learning Engineer - NLP (Internship) Apr 2021 – Oct 2021
Contactel - Novelis Paris, France
Built a **classification tool for customer exchange archives** for an energy company, navigating shifting class distributions and scarce labeled data.

- Explored **TF-IDF + XGBoost** (**98% accuracy**) and **XLNet** fine-tuning (**96% accuracy**) but limited by continuously evolving classes
- Adopted **BERT sentence embedding + similarity** as the final solution for its label-free generalization to new classes
- Performed **topic modeling (LDA)** for cluster exploration and visualization
- Deployed as a **REST API** via Flask + Docker on GCP

Stack: Python, PyTorch, BERT, XLNet, XGBoost, Gensim, Flask, Docker, GCP

Machine Learning Engineer - Computer Vision (Internship) Nov 2020 – Mar 2021
CRIStAL Labs Lille, France
Defined and deployed a **Detectron2 (Mask R-CNN)** obstacle detection model on a **Renault ZOE** autonomous vehicle.
- Adapted to **ROS**, compiled with **TorchScript**, deployed on **NVIDIA Jetson**

Stack: Python, PyTorch, Detectron2, TorchScript, ROS, NVIDIA Jetson, Linux

Projects

Debby Aug 2025 – Dec 2025
Founder & Software/ AI Engineer
Solo-built platform analyzing content (videos, PDFs, articles, tweets) for credibility, fact-checking, bias, fallacy detection.
- Architected a **modular multi-stage AI pipeline** with specialized engines for ingestion and analysis
- Implemented **credit-based billing** (Polar), token-level cost tracking, and workpool job queuing
- Integrated **Langfuse** (LLM observability) and **PostHog** (analytics)

Stack: Next.js 15, TypeScript, Convex, Anthropic, Gemini, Vercel AI SDK, Langfuse, Cloudflare R2

Bitmatcha Dec 2025 – Jan 2026
Founder & Software/ AI Engineer
Solo-directed, **Cursor-assisted** full-stack platform matching developers with cofounders via GitHub activity.
- Designed a **custom scoring algorithm** using GitHub GraphQL API (commit activity, skills, location)
- Built real-time **1:1 chat** with read receipts and paginated match lists
- Shipped in weeks

Stack: Next.js 15, TypeScript, Convex, GitHub OAuth, Resend, PostHog

Ardoise Sep 2024 – Jul 2025
CTO & Cofounder
Co-founded a SaaS platform transforming documents into interactive quizzes for educators and HR teams.
- Built **AI-powered document parsing** with real-time streaming for quiz generation
- Set up **CI/CD** (GitHub Actions), Docker, semantic versioning, Sentry

- Integrated **Stripe** billing and OAuth authentication

Stack: Next.js 14, TypeScript, Tailwind, Stripe, Docker, GitHub Actions, Sentry

Languages

English ● ● ● ● ● **French** ● ● ● ● ●
TOEIC 970

Arabic ● ● ● ● ●

Certificates

MOOC Project Management Advanced Track + Specialties

TensorFlow Developer Certificate

Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization

Sample-based Learning Methods

Fundamentals of Reinforcement Learning

DeepLearning.AI TensorFlow Developer

Sequences, Time Series and Prediction

Natural Language Processing in TensorFlow

Convolutional Neural Networks in TensorFlow

Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning

Machine Learning

Probability and Statistics: To p or not to p?